

Food Frontiers and Security Science Program

Inception Report

June 2025

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Acronyms

ACIAR	Australian Centre for International Agricultural Research
AI	Artificial Intelligence
ANCORS	Australian National Centre for Ocean Resources and Security
AoW	Area of Work
AWARE	Early Warning, Early Action and Early Finance
CA	Comparative Advantage
CIAT	International Center for Tropical Agriculture
CIP	International Potato Center
CWANA	West Asia and North Africa
EA	Evaluability Assessment
ESA	Eastern and Southern Africa
FAO	Food and Agricultural Organization of the United Nations
FCAS	Fragile and conflict-affected situations
FCDO	Foreign, Commonwealth & Development Office (United Kingdom)
FCM	Fragility, Conflict and Migration
FLEX	Frontlines Learning Exchange
HLO	High Level Outcome
IAES	Independent Advisory and Evaluation Service
IFPRI	International Food Policy Research Institute
IITA	International Institute of Tropical Agriculture
IFRC	International Federation of Red Cross and Red Crescent Societies
ILRI	International Livestock Research Institute
IPSR	Innovation Package and Scaling Readiness
IRRI	International Rice Research Institute
ISDC	Independent Science for Development Council
IWMI	International Water Management Institute
LAC	Latin America & the Caribbean
MEL	Monitoring Evaluation Learning
MELIA	Monitoring, Evaluation, Learning, and Impact Assessment
MFMR	Ministry of Fisheries and Marine Resources (Solomon Islands)
PMU	Program Management Unit
PORB	Plan of Results and Budget
PRMS	Program Results Monitoring System
SA	South Asia
SDC	Swiss Agency for Development and Cooperation
SEA	Southeast Asia
SP	Science Program
SPC	The Pacific Community
UN	United Nations
UNHCR	United Nations High Commissioner for Refugees
USD	United States Dollar
WCA	West and Central Africa
WFP	United Nations World Food Programme
WVI	World Vision International

Executive summary

The [CGIAR Food Frontiers and Security Science Program](#) represents an exciting new direction in the CGIAR Portfolio. The Program is co-designed and engages with three frontier food systems:

- Fragile and conflict-affected situations (FCAS) – Area of Work (AoW) 2
- Urban and peri-urban food systems – AoW3
- Island food systems – AoW4

These systems, while distinct, share many common challenges and represent high-leverage entry points for innovation and resilience-building solutions (Figure 1). Frontier food systems are emblematic of the Anthropocene, where intersecting risks challenge traditional development approaches but also unlock opportunities for systems transformation. Complementing the above three AoWs, the Future Food Systems Lab (AoW1) brings the three together, serving as a global hub for synthesis, foresight, and innovation across frontier food systems—connecting local experimentation with global policy relevance.

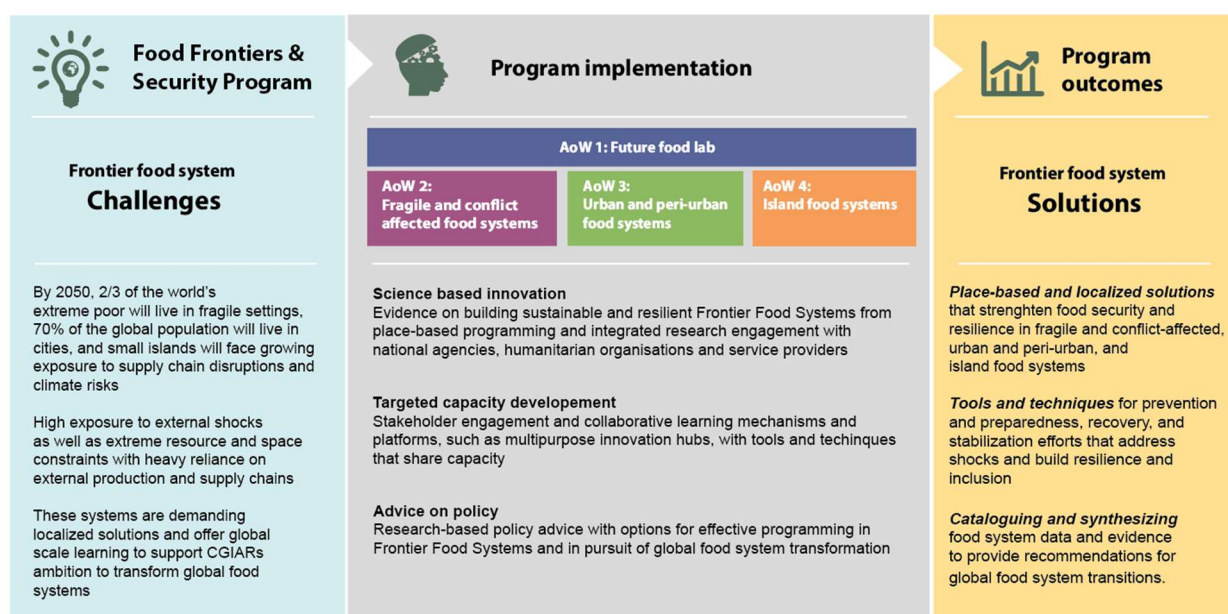


Figure 1. Food Frontiers and Security Program logic outlines the Science Program structure from challenges to solutions.

During the [2025-30 CGIAR Research Portfolio](#) Inception Phase (January-June 2025), the Food Frontiers and Security Program analyzed its Comparative Advantage and found that CGIAR holds an important role in frontier geographies due to its embedded partnerships, innovation capacity, and systems research approach (Section 4). This allows CGIAR to convene key stakeholders, bridge policy and evidence gaps, and co-develop solutions to complex challenges. This analysis has clarified CGIAR's value added and refined its strategic focus.

The knowledge-driven Prioritization in this report exercise further reinforced the Program's emphasis on Eastern and Southern Africa, South Asia, Southeast Asia and the Pacific (as the only Program with a dedicated emphasis in this region), identifying these as regions where frontier systems challenges are particularly acute and CGIAR can deliver high-impact, integrated research and innovation. Programmatic choices have been guided not only by Prioritization, but also by donor interest, partner demand, and CGIAR's institutional readiness (Section 5).

The Food Frontiers and Security Program's Theory of Change (ToC) has been refined to align with a systemic approach to impact—one that is co-created, non-linear, and rooted in institutional change, anticipatory action, and informed interventions and programming (Section 3). Its

Monitoring, Evaluation, Learning, and Impact Assessment (MELIA) Plan supports adaptive management and accountability (Section 3.2).

The Program has successfully managed an uncertain funding landscape, prioritizing flexibility, integration with bilateral efforts, and co-financing with partners (Section 9). Strategic budgeting and contingency planning help safeguard delivery in fragile and dynamic settings, as outlined in the Plan of Results and Budget (PORB) (Section 7). Key partnerships across the portfolio ensure that the Program is well integrated and engages in system-wide collaboration on scaling, policy evidence generation, and innovation. Cross-portfolio and regional coordination mechanisms through CGIAR structures further enhance coherence, efficiency, and impact (Section 8).

The Food Frontiers and Security Program enters the implementation phase with a bold, clear, and adaptive strategy to generate actionable research and drive inclusive transformation—positioning CGIAR to contribute meaningfully to global efforts to address the intersecting crises of climate change; fragility and conflict; and food insecurity. In an era defined by the Anthropocene, the Program offers the CGIAR a timely vehicle to build resilience and equity in systems most at risk and overlooked by mainstream innovation and science.

1. Program overview

1.1 High-level Vision

The Food Frontiers and Security Program seeks resilient and inclusive food systems in frontier settings as critical levers for global climate change adaptation and food system transformation in the Anthropocene. “Frontier” food systems are those situated at the edges of current global research, investment, and innovation priorities—geographically and institutionally. We focus on three frontier food systems: (i) FCAS, (ii) urban and peri-urban environments experiencing urbanization and rapid transformation, and (iii) small island states and territories. All three are characterized by compounded vulnerabilities—including institutional fragility, environmental and spatial constraints, production constraints, and exposure to intersecting shocks such as conflict, climate extremes, displacement, and rapid dietary transitions. These characteristics distinguish them from more stable, self-sufficient, and agriculturally dominant ones (Figure 2).



Figure 2. Satellite images of Malé (left), the capital of the Maldives, and Zaatari refugee camp in Jordan (right) help illustrate analogous systems of urban, island and fragile and conflict-affected food systems. They are tethered worlds often relying on outside input to compensate for lack of space and resources, in turn experiencing exposure and vulnerability to similar shocks and risks.

Critically, frontier food systems are not peripheral in importance: they are central to achieving resilience, equity, and climate adaptation globally, despite having historically fallen outside the core focus of agricultural research and development. Transforming global food systems has become a top priority in the [2030 Agenda for Sustainable Development](#). Trends of rising urbanization, population growth, and political instability occurring in frontier food systems demand cutting-edge research and innovation that address such challenges and drive CGIAR’s work into the future. These systems represent innovation niches with massive potential for learning, scaling, and impact amid declining aid budgets.

This Program is supporting frontier systems as key examples of the broader global transformation challenge. It focuses on climate risk-informed, context-specific, and transdisciplinary approaches to strengthen food, land, and water systems in these complex and often fragile settings. During the Inception Phase, the Program developed a clear logic for its engagement with these three frontier systems, refining its pathways to impact in collaboration with local partners. The Program’s applied research agenda is explicitly designed to inform both development and humanitarian responses, recognizing that resilience-building in frontier systems requires bridging the humanitarian-development-peace nexus and addressing core trends that requires more than CGIAR science-as-usual.

1.2 Achievements from 2022—2024

The Food Frontiers and Security Program brings together—and expands on—achievements from three CGIAR Initiatives in the 2022-2024 Portfolio: [Fragility, Conflict, and Migration \(FCM\)](#), [Resilient Cities](#), and [Aquatic Foods](#). These laid critical foundations for the 2025–2030 Portfolio by generating actionable research and forging transformative partnerships across more than twelve fragile and conflict-affected countries; building resilience of food, land, and water systems in rapidly growing cities in Africa, Asia, and Latin America; and pioneering a new framework for innovation and scaling in island systems (particularly in the Pacific). As an example, FCM informed 41 policy changes related to anticipatory action, emergency response, and recovery and resilience programming between 2022-2024 which we will build and expand upon.

The design of this new Science Program (SP) builds on these foundations while advancing a new generation of thinking and action that explicitly integrates climate adaptation and mitigation strategies and addresses undeniable surges in crises, conflict, and urbanization. This includes a novel framework for research and innovation transfer and integration of learning from local innovation hubs (e.g. [Quezon Vendor Business School](#), [Nusatupe Innovation hub](#) for Island Food Systems, [Ghana's Circular Bioeconomy Hub](#)). These efforts are further amplified through planetary scale learning facilitated by the Future Food System Lab, which bridges science-policy engagement and locally led innovation to address intersecting risks of climate change, conflict, and food insecurity.

1.3 Description of intended research

Value proposition The Program's value lies in generating empirical evidence that drives both local action—across development and humanitarian contexts—and global guidance on food system transformation. From the outset, the 2025–2030 Portfolio emphasized the importance of transdisciplinary research in frontier geographies that delivers local impact while informing global solutions. By 2030, two-thirds of the world's extreme poor will live in FCAS; by 2050, nearly 70 percent of the global population will reside in urban areas, and societies in 57 Small Island Developing States are facing unprecedented food system challenges. Addressing the challenges faced by these communities through science-based solutions is central to the Program's mission.

Research on food systems often veers between overly global generalizations and narrowly local studies. Yet, many key challenges—like climate change, supply chain disruptions, and institutional fragility—occur at intermediate scales. The Food Frontiers and Security Program fills this gap by conducting comparative, transdisciplinary research across thematically linked geographies. This meso-level approach reveals generalizable insights rooted in local realities, producing scalable, actionable knowledge that strengthens food system resilience and delivers international public goods.

The Program addresses a core set of research priorities that reflect the interconnected challenges and structural drivers of food system transformation in fragile, urban, and island settings. Examples of common research topics include: (i) scenario planning to anticipate emerging risks and inform strategic responses; (ii) system characterization, synthesis and adaptive learning to support iterative program design and decision-making; (iii) identification and scaling of innovations, such as nature-based, circular, and digital solutions; (iv) inclusive development, with a strong emphasis on engaging women, youth, and marginalized groups in co-creating and delivering solutions. By aligning research design, implementation, and cross-context learning, the program deepens understanding of how integrated, inclusive, and adaptive strategies can enhance resilience in frontier food systems.

Research strategy The Program is a transdisciplinary research enterprise, and each Area of Work (AoW) includes research topics requiring distinct methods and approaches, as outlined below. We will make our research impactful through integration within positive development trajectories, carefully prioritizing not only *what* we do, but also *how* we work. For example, High Level Outcome (HLO) 1.3 brings together CGIAR researchers, humanitarian and development actors, governments, private-sector innovators, and communities to collectively address the question: Which agrifood innovations best enhance resilience and livelihoods in frontier areas, and how can their scaling be accelerated? This integration of mixed-methods research, innovation practice, and policy engagement exemplifies the program’s transdisciplinary approach to addressing complex challenges in frontier geographies.

Approaches For the Program to be successful in its ambition, it is important to be deliberate about integration with the national structures and coalitions in which teams are operating. The consortium of institutions implementing this Program has a strong track record of building and strengthening impactful research partnerships, evidence, and innovations (see [Section 2](#)). Program teams will be working in different ways depending on the scope of their research, but all share a common focus on ensuring relevance and legitimacy as essential foundations for research effectiveness. For example, amid global reductions in humanitarian and research funding, the Program is innovating with low cost, high impact integration approaches to applied science through staff secondments - embedding research staff directly in humanitarian and development agencies. This approach is well-suited to some circumstances and partnerships, while in others, different approaches—including periodic engagement and co-creation of evidence—are being pursued.

Methods Diverse modalities of engagement and partnership are also reflected in the choice of methods. For example, larger scale (e.g. at the scale of island or city) food systems’ resilience assessments rely largely on analyses of public domain data. For this kind of work, analysts operationalize co-designed research frameworks to generate assessments that are validated through periodic reflections and engagements within partnerships. For research activities that require field data, more engaged individual-level and farm-level analyses using a mix of quantitative and qualitative research methods are employed. Some examples of employed methods within the Food Frontiers and Security Program research themes are tabulated below.

Research Theme	Method
Anticipatory action programming	Scenario modelling, risk forecasting, early warning analyses
Interventions for crisis response and recovery	Randomized control trials, quasi-experimental econometric analysis, qualitative data collection, meta- analysis and synthesis of existing research and evidence
Women’s empowerment	Quantitative/ econometric analysis of new and existing survey data
Migration	Quantitative/ econometric analysis of new and existing survey data
Bio circularity and Circular economy	Life-cycle assessments, Nutrient flow analyses, chemical water analyses
Climate risk	Compound risk and impact analysis targeted to food systems vulnerabilities and its population (e.g. refugees, small island and urban population)
Island food system resilience	Food trade analyses, food production estimations, consumption analysis, dietary diversity analyses, systematic literature review.

Food safety, food waste and loss	Organoleptic quality assessment, gender-responsive food-loss assessment
Peace and security nexus	Political economy analyses, climate, and conflict sensitivity analysis, social network analyses

The Food Frontiers and Security Program breaks new ground by piloting science-policy co-creation models, deploying low-cost digital monitoring tools in data-scarce areas, working with partners to maximize reach and operating in challenging settings where new in-person data collection is needed, and embedding researchers in frontline institutions. Expected outputs include science-driven evidence on optimal programming, policies, and interventions to anticipate and respond to shocks and crises, as well as to stabilize rural livelihoods and recovery; open-access policy tools; foresight scenarios; food system diagnostics; and operational guidance for scaling context-specific innovations. Across all frontier systems, the Program will deliver actionable evidence and tools that improve food, land, and water system resilience; inform programming, investments, and governance to address frontier challenges before, during, and after they arise; and foster locally-led and gender-sensitive innovation in the face of systemic risks.

2. Codesign and partnerships

Recognizing that innovation in frontier systems demands legitimacy, agility, and grounded engagement, the Program has deliberately prioritized partnerships that foster contextual relevance, institutional integration, and long-term impact potential.

2.1 Inception phase engagement

Throughout the Program's design and inception phases (2024–June 2025), diverse engagement mechanisms were deployed to go beyond consultation and foster shared ownership of the design process:

- [CGIAR Science and Partnership Week](#) (April 7-11, 2025) convened over 4,500 attendees from 100 countries. The Food Frontiers and Security Program convened three strategic dialogues on frontier food systems science, fragility and conflict, and urban and island innovations. These sessions engaged a large array of Program partners working in fragile and conflict-affected, urban and peri-urban, and island food systems and have directly shaped the research agendas of the AoWs. For example, for AoW1, these sessions reinforced the demand for integrating systems and foresight analysis in the planning and synthesized evidence on effective program delivery strategies.
- The [Sudan Conflict Conference](#) (April 14-15, 2025), held in-person in Nairobi, enabled regional actors and humanitarian organizations to deliberate on effective humanitarian response, which contributed to shaping AoW2's priorities in Sudan, particularly on effective responses and modalities of humanitarian response under protracted crises. Identified knowledge gaps have been incorporated into ongoing work and MELIA plans in Sudan (see Section 3) and will be incorporated into a book being prepared during 2025 for 2026 publication.
- From [Relief to Resilience in Somalia: Harnessing Research to Inform Social Protection Policy and Practice](#) (June 2025) will serve as a public showcase of the insights and priorities emerging from a series of smaller dialogues. The event will highlight Somalia's shift from short-term aid to long-term development through integrated social protection strategies. Government officials, partners, and researchers will share lessons on supporting internally displaced people and reducing poverty, which will directly inform the development of a learning agenda that the Program will take up for Somalia.
- A global webinar on [Anticipatory Action Toolkit](#) engaged over 100 stakeholders from humanitarian and development sectors, offering practical feedback to refine program outputs and further science-policy coordination.
- We consulted local innovation partners, including Pacific partners at the [Nusatupe Innovation Hub](#) for island food systems and the [Quezon Vendor Business School](#), who contributed directly to the design of innovation scaling models—offering a blueprint for place-based delivery in other geographies.
- During the Inception Phase, the Food Frontiers and Security Program engaged extensively with other CGIAR SPs and Accelerators through joint workshops to co-design activities, identify synergies, and ensure alignment.

Each AoW was co-developed with regional and thematic partners—including United Nations (UN) agencies, local governments, grassroots innovation hubs, and national ministries—to ensure alignment with real-world, demand-driven priorities and policy and implementation entry points. For instance, AoW1 emerged in response to stakeholder calls for a global platform connecting frontier innovations with decision-making processes.

Similarly, AoW2 on Fragile and Conflict-Affected Food Systems was shaped through consultations during the 2025 dialogues with partners like the [United Nations World Food Programme \(WFP\)](#), [United Nations High Commissioner for Refugees \(UNHCR\)](#), [International Organization for Migration \(IOM\)](#), [ActionAid International](#), and [World Vision International \(WVI\)](#) to design inclusive, conflict-sensitive strategies that bridge humanitarian action and long-term development. Across the Program, ToCs were revised to reflect local dynamics and institutional contexts, while timelines and deliverables were aligned with national policy cycles and partner planning windows.

2.2 Pursuing new and innovative ways of working

In the CGIAR 2025-2030 Portfolio, “Frontiers” refers to both emerging geographies and innovative approaches that challenge traditional development research. Legitimacy is regarded as key to effectiveness, with co-created, and relevant research being essential for trust and uptake in fragile or underserved systems. The Future Food Systems Lab (AoW1) integrates research across AoWs and frontier systems, generating insights that are both generalizable and locally grounded. Staff secondment that sees CGIAR researchers embedded within humanitarian agencies, ministries, and non-governmental organizations (NGOs) offer a low-cost, context responsive model for real-time science delivery. These co-developed secondment models and formal links with bilateral projects have been institutionalized in the design of AoWs, therefore expanding the Program’s research scope, partnerships, and innovations.

2.3 Adaptive management through structured reflection processes

To stay responsive to emerging risks and evolving partner needs, the Program uses CGIAR’s institutionalized Reflect → Re-plan cycle. This formal mechanism enables joint reflection with partners to assess progress, update priorities, and revise key elements such as the Plan of Results and Budget (PORB), ToC, and risk register. It also promotes transparency by producing a one-page public “Reflect write-up,” approved by the Program Director and incorporated into the annual technical report. In 2025, the reflect process will pilot inclusive formats involving AoW leads, co-leads, local partners, and MELIA teams to integrate partner-generated evidence and user feedback into strategic adjustments. Reflective practice is further institutionalized through learning engagements and webinar series such as the Frontlines Learning Exchange (FLEX) encompassing AoW1’s synthesis work as well as AoW2’s ‘From Fragility to Stability’ webinar series. By offering structured platforms for collaboration that allow evidence sharing and peer learning, we ensure the Program stays adaptive and responsive to dynamic, complex FCAS.

2.4 Forward-looking co-design mechanisms

Co-design will continue to define the Program’s implementation phase, supported by forward-looking mechanisms to sustain that inclusive engagement and shared ownership. Each AoW will host annual “Signature Moments”—including policy launches, learning briefs and evidence campaigns co-hosted with partners to build momentum and foster co-ownership. Regular engagement will be supported through newsletters, webinars, and curated policy updates, fostering transparency and joint learning. Additionally, foresight sprints and scenario workshops—led by AoW1 and regional hubs—will respond to major shocks, such as conflict or climate events while supporting long-term planning processes. Digital collaboration tools, including monthly update calls and shared content repositories, will enable real-time coordination and knowledge exchange across CGIAR and its global partner network.

Looking ahead, the Food Frontiers and Security Program will continue to collaborate closely with international organizations and NGOs, the private sector, and public sector partners. The Program’s impact pathways are designed with and through our partners—such that remaining in close collaboration with them is critical for the Program’s impact and long-term success.

3. Theory of change and MELIA

3.1 *Theory of change*

The Food Frontiers and Security Program's ToC ([diagram](#)) articulates a clear and actionable framework, directly linking Program activities (outputs) to anticipated changes (outcomes) and ultimately, to the desired overarching goal of enhanced resilience in frontier food systems (impact).

The Program's initial outputs include developing and disseminating tools, innovations, and actionable resilience-building plans to CGIAR partners and funders. This leads directly to the outcome of key stakeholders being better prepared to anticipate emerging risks—including those related to climate, instability, conflict, and health—and make strategic decisions that foster long-term sustainability within frontier food systems. Simultaneously, the Program produces outputs such as tailored training, decision-support tools, and business model development for small and medium enterprises (SMEs), local innovators, and community-based organizations (CBOs). The direct outcome of these interventions is the empowerment of food system actors to effectively scale context-specific innovative solutions, which in turn contributes to the impact of improved livelihoods, poverty reduction, women's empowerment, better health and nutrition. The Program generates crucial information that enables the outcome of policymakers and key decision-makers crafting more adaptive and inclusive (including otherwise socially, economically, and politically excluded groups encompassing women, youth, and Indigenous people) policies and programming that are responsive to the unique vulnerabilities of frontier systems.

The Program generates vital information contributing to inclusivity. Our outputs include evidence on gender- and other vulnerability-disaggregated impacts, context-specific vulnerability assessments, and best policies, programming, interventions, and innovations for gender-responsive and socially inclusive solutions. The mechanism connecting this information to policymakers creating inclusive policies is direct engagement. We facilitate tailored policy dialogues and workshops, offer decision-support tools with training, and invest in capacity-sharing initiatives. This rigorous, localized evidence builds trust and demonstrates the tangible benefits of inclusive approaches, directly enabling decision-makers to craft more equitable and responsive policies that address the unique needs of excluded groups. Further, the Program's output also includes investing in capacity-sharing initiatives, which shift power and foster genuine local leadership. The resulting outcome is strengthened local agency and ownership over resilience-building efforts, ensuring that innovations are both locally driven and sustainable.

The Program's success hinges on key assumptions. We assume diverse actors, ranging from policymakers to local innovators, will willingly apply our evidence and tools. If not, our policy guidance (output) will not necessarily lead to effective policies (outcome), and local training (output) will not scale solutions—thus failing to improve livelihoods. We also assume that strong, collaborative partnerships will be in place to support uptake and scaling across varied contexts. We will actively monitor these assumptions, mitigating risks via co-creation, demonstrating tangible benefits and capacity-sharing to encourage uptake and adapt strategies.

The Future Food System Lab (AoW1) functions as the Program's integrative engine - synthesizing evidence and fostering reciprocal learning across fragile, urban, and island food systems. Guided by the unified program logic outlined in Section 1.1, AoW1 identifies cross-cutting challenges and transferable insights across these diverse frontier geographies. This synthesis draws on literature reviews, meta-analyses, primary biophysical and socio-economic data, case studies, participatory workshops, and stakeholder consultations. Through this process, AoW1 generates comparative

lessons, practical frameworks, and decision-support tools that inform adaptive strategies across AoW2, AoW3, and AoW4. In doing so, it ensures continuous refinement of program activities through an integrated systems lens, enhancing the coherence and impact of food system transformation efforts in resource-constrained settings.

During the Inception Phase, we refined our ToC. These updates sharpened the Program’s logic, clarity, and pathways to impact. HLOs and Intermediate Outcomes (IOs) were a primary focus of the revisions, with changes made to clarify distinctions, improve differentiation, and align language to maintain consistency across AoWs. The Program team worked within their AoWs to streamline and align deliverables and targets across the Program to enhance coherence, feasibility, and responsiveness to local needs. Intermediate outcomes were revised to better reflect the full scope of work and allow for future innovation. For example, the [country program in Solomon Islands](#) now adopts the island food system framework (Figure 3).

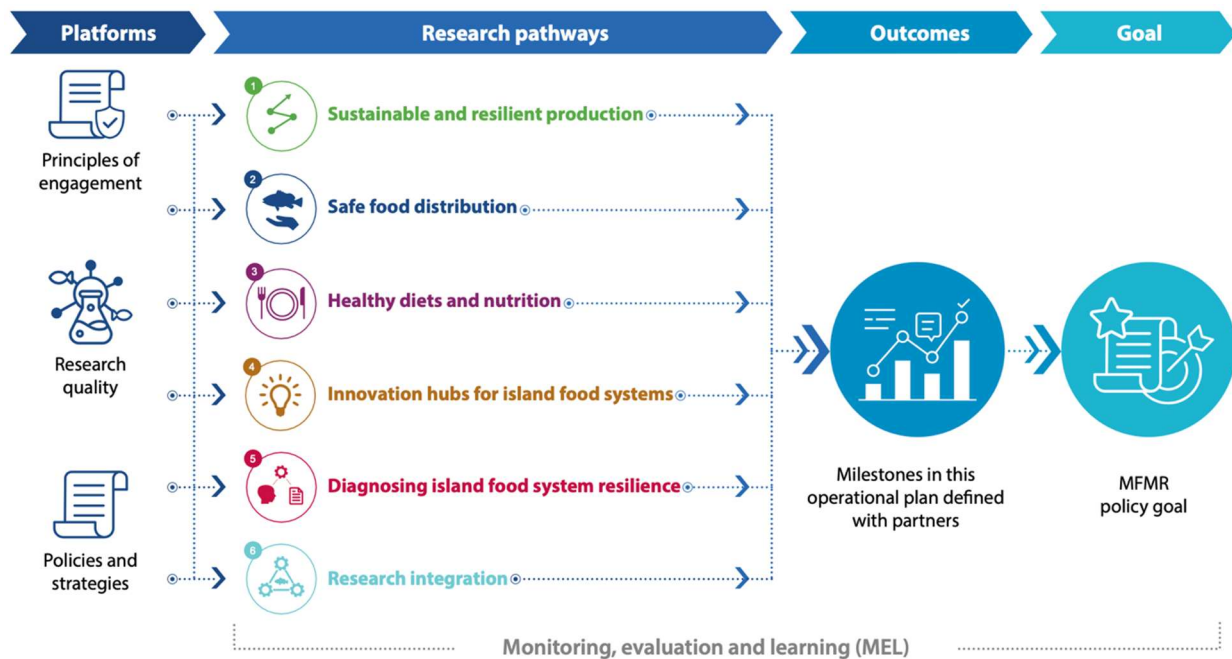


Figure 3. The WorldFish and CGIAR engagement in Solomon Islands adopts the six components of the island food system framework developed in the Food Frontiers and Security Science Program informing the Ministry of Fisheries and Marine Resources (Solomon Islands).

Across the Program, teams sought partner inputs—through workshops, consultations, and existing partner networks—to validate targets, indicators, and investment priorities (see Section 2). These efforts ensured that activities were grounded in their context, tailored to the realities of frontier food system settings, and aligned with both stakeholder needs and the overarching ToC. Integration with ongoing bilateral projects further strengthened the Program’s foundations and practical relevance from the outset.

The Program used feedback from the Independent Science for Development Council (ISDC) to sharpen its ToC (see Appendix 1). Here, AoW responded uniquely: AoW1 strengthened the “Future Food System Lab” for cross-geography learning, clarifying output-outcome links and refining assumptions. AoW2 reduced its HLOs from five to three, prioritizing specificity and integrating the ToC into Program management. AoW3 enhanced implementation, scaling, and co-design with partners, setting clearer targets. Finally, AoW4 improved its ToC by increasing the specificity of outputs and assumptions, detailing what each entail and how they differ, while also refining causal mechanisms by clearly outlining research-to-impact pathways and the roles of partners at each

step. These collective efforts underscore the Program’s ambition and commitment to impactful work.

Some parts of the Food Frontiers and Security Program’s ToC were also directly influenced by the Independent Advisory and Evaluation Service’s (IAES’s) evaluation findings, leading to clearer logic, sharper assumptions, and more targeted impact pathways across all AoWs. Furthermore, a key evaluation insight pushed for more collaboration across SPs to address global polycrises, which resulted in new collaborations, including with the Policy Innovation SP (see Section 8.1). While updating the ToC, we also addressed the informal feedback received from the Evaluability Assessment (EA) team, particularly their advice to be strategic in indicator selection to manage cost and complexity, and to ensure clarity in the ToCs logical flow, HLOs, and IOs when building our MELIA Plan.

Achieving resilience in frontier systems demands a systemic, co-created, and adaptive approach, not isolated interventions. We do this through continuous engagement with SP leaders via biweekly meetings, ensuring alignment and leveraging expertise. To channel insights from synthesis work back to AoWs2, 3, and 4, AoW1 convenes cross-AoW working groups on key themes (such as resilience analysis and entrepreneurship support) as well as monthly coordination meetings to facilitate continuous exchange and alignment. Our highly integrated partnership strategy involves sharing our ToC to foster shared understanding and actively engaging other SPs to expand collective reach (e.g., we have joined the Policy Innovation SP to organize an event on crisis in Sudan, and are organizing a session on gender and fragility at the upcoming [CGIAR Gender Conference](#) in October 2025, as well as co-investing with the [Genebanks](#) on managing genetic diversity in the Pacific).

As the Food Frontiers and Security Program’s leadership, we have drafted a paper that will be published on the strategic rationale for collectively addressing food systems across diverse frontier geographies—fragile, urban, and island contexts—to generate transferable insights through a unified analytical framework. In terms of funding, while our core ToC components benefit from pooled funding, at present, only bilateral projects linking to work under AoW2 and AoW3 have been mapped to the Program (see Section 6). AoW4 garnered bilateral projects that are below the 80 percent threshold for Center reporting in 2025. Most of the bilateral project funding is under AoW2, supporting all three of its outcome areas: preventing, responding to, and recovering from shocks and crises.

3.2 Monitoring, Evaluation, Learning, and Impact Assessment (MELIA)

The Food Frontiers and Security Program’s MELIA Plan incorporates Science Group and EA team feedback for effective deliverables tracking, and adaptive learning across all AoWs. Following IAES guidance, indicators were chosen for appropriateness, cost-effectiveness, and data quality. We also ensured dedicated Monitoring Evaluation Learning (MEL) support personnel and baseline evidence for each intervention. This function will be centrally managed by the Program Management Unit (PMU), working in close partnership with AoW leads and Center Delegates.

The Program has a comprehensive monitoring strategy and tool which is currently under development and is being trialed by AoW leads. Once finalized, the AoW leadership will be responsible for ensuring deliverables, including knowledge products, capacity development, and innovation development results are tracked, and (with all supporting evidence) saved in CGSpace or other publicly available fora, in line with CGIAR’s technical reporting guidelines and quality assurance mechanisms. The tracking tool will be completed directly by each Center and saved in shared folders, enabling real-time, robust Program-level progress tracking. For outcomes, we will employ a multi-faceted approach, such as surveys and interviews. Innovation use will be monitored via midline surveys, detailed logs on adoption and reach, qualitative case studies, and catalogued

in CGIAR innovation reporting tool. Policy influence will be tracked through mentions in government documents, the media, policymaker interviews, and written correspondence. Impacts on food system actors and livelihoods will be assessed through midline/endline surveys, adoption tracking, and qualitative data from focus groups and success stories. Data will be collected regularly based on each indicator's reporting requirements. AoW leads will monitor and ensure the quality of data reported in the Program Results Monitoring System (PRMS).

The MELIA Plan includes a robust evaluation component to understand Program effectiveness and inform adaptations. AoW1 will carry out case studies on vendor business models. AoW2's evaluations will assess the impacts of interventions including those related to social protection, livelihoods programming, disaster response, and crisis preparedness in fragile settings, including analyzing gendered impacts. Surveys and ex-post studies will be conducted to assess Program design and results. AoW3 will evaluate the impact of training urban dwellers in growing hardy, nutrient-dense vegetables in urban gardens on diet quality, while AoW4 focuses on the impact of improved aquatic food handling and Indigenous food promotion. These initial studies are supported by pooled funds, with further Program-level evaluations contingent on budget. Details on evaluations within the bilateral projects have yet to be collected.

By 2030, causal impact assessments (IAs) will prioritize novel interventions in under-researched frontier contexts. These assessments will evaluate short/medium-term CGIAR Impact Area impacts, testing ToC assumptions to inform future programming. For example, we plan a qualitative evaluation to assess knowledge uptake and beneficiary reach under AoW2, with similar studies anticipated for other AoWs as budget allows. All relevant quantitative data will be disaggregated by sex and age. Qualitatively, participatory approaches and contextualized case studies will illuminate how the Program makes a nuanced difference. Annual "Pause & Reflect" events will enhance adaptability and refine Program design (see Section 2.3).

Refer to the MELIA Plan and Results Framework attachments for additional details.

4. Comparative advantage analysis

In underserved and often isolated frontier geographies, CGIAR is uniquely positioned to contribute. The Comparative Advantage (CA) analysis across all AoWs highlights CGIAR's capacity to deliver high-level outputs and outcomes by leveraging its research for development expertise and partnerships. In the Food Frontiers and Security Program, CGIAR supports communities in these regions to drive progress across all Impact Areas and build resilience. Our CA analysis highlighted the diverse strengths of various partners, such as [WFP](#), [UNHCR](#), the [World Vegetable Center](#), the [World Bank Group](#), the [Food and Agricultural Organization \(FAO\)](#), the [International Federation of Red Cross and Red Crescent Societies \(IFRC\)](#), [CARE](#), [WVI](#), and [ActionAid International](#), as well as national and regional governments and other regional and country-specific actors like [Sahel Consulting](#), [Red Cross Red Crescent Climate Centre](#), [Kastom Gaden Association](#), [The Pacific Community \(SPC\)](#), [Solomon Islands National University](#), [Addis Ababa Municipality](#), and [Quezon City Government](#). These entities offer complementary strengths in local knowledge, operational capacity, policy influence, and community engagement capabilities, forming a strong foundation for impactful collaboration.

CGIAR complements this network by bringing its unique expertise in applied research, program evaluation, innovation, and scaling science. By developing tools, methods, data, and approaches for complex frontier contexts, CGIAR enhances the transformative potential of locally developed successes. Its scaling capabilities ensure that these innovations are not only applied effectively and adapted across similar geographies, amplifying impact. CGIAR and its partners form a dynamic alliance that leverages shared strengths to deliver targeted solutions. For example, AoW3 activities bridge the gap between international research (conducted in strategically selected geographies representing urban food system challenges) and grassroots implementation, translating global insights into local solutions. This feeds into global synthesis under AoW1, which documents systemic challenges and identifies transformative pathways for resilient urban food systems – maintaining and strengthening CGIAR's CA internationally.

The CA analysis also clarified the complementary roles across the portfolio. CGIAR leads diagnostics, evidence generation, and innovation design; partners focus on delivery, policy uptake, and scaling. For example, in FCAS, CGIAR's scientific rigor pairs with the operational reach of humanitarian actors and the contextual expertise of national and subnational institutions. This division supports a collaborative ecosystem where each actor contributes uniquely.

CGIAR ensures a pipeline of evidence-based solutions, while partners with local networks accelerate their practical application—making interventions more context-sensitive, scalable, and sustainable. This synergy extends to policy advocacy and capacity sharing. CGIAR produces research and insights to collaborate with partners, turning collective knowledge into actionable policies and solutions. Together strengthening local institutional and community practices—an essential step toward long-term resilience. This collaborative model ensures Program innovations are technical sound, inclusive, grounded, and impactful. Importantly, the CA analysis influenced prioritization of geographies, activities, and partnerships. Activities now focus on countries where CGIAR has strong relationships and high demand for co-developed tools. The analysis also informed decisions to phase out or reposition workstreams where CGIAR's CA was weaker. For example, five AoW2 HLOs were consolidated into three (Prevent, Respond, Recover). In AoW4, CGIAR identified an opportunity to leverage its advantage in cataloging and managing genetic diversity in the Pacific, leading to collaboration with [Genebanks](#).

Overall, this CA-informed approach has sharpened the strategic coherence of the Program. It aligns research priorities with partner capabilities and highlights CGIAR's role in bridging science and action across the frontiers. Examining these systems together has revealed common challenges—environmental degradation, resource scarcity, institutional fragility, and vulnerability to climate shocks—as well as shared opportunities for innovation, learning, and scaling.

Refer to the Comparative Advantage attachment for additional detail.

5. Prioritization

5.1 Knowledge-driven priorities

The Prioritization clearly highlights a strategic focus on Eastern and Southern Africa (ESA), South Asia (SA), and Southeast Asia (SEA), which includes the Pacific. These regions consistently rank high across all five impact areas. They show strong potential for integrated programming, especially where resilience, inclusion, and urban food system interventions intersect. West and Central Africa (WCA) also emerges as a key geography, especially for food security, poverty reduction, and climate-related interventions.

Across the impact areas, certain HLOs and AoWs emerge as cross-cutting priorities. These include: HLO1.2 (co-developing action plans and innovation priorities across the frontier geographies), AoW2.2 (humanitarian response effectiveness), and AoW3 (urban food production systems, circular bioeconomy, informal markets, and healthy diets). In the Pacific (AoW4), gender-sensitive and youth-oriented tools, island food system networks and capacity-building, show particularly high scores, pointing to unique regional opportunities for food, land, and water systems innovation in island contexts.

This assessment underscores the importance of engaging with geographies with frontier food systems like ESA, SA, SEA and the Pacific, where resilience-building efforts can effectively integrate food security, climate adaptation, and social inclusion through innovative programming. By prioritizing high-impact HLOs across AoWs, such as food security, poverty reduction, and gender-sensitive programming and innovations, the analysis aims to strengthen the adaptive capacities of vulnerable populations while fostering sustainable development.

By contrast, Central & West Asia and North Africa (CWANA) and Latin America & the Caribbean (LAC) appear as lower priorities, with few HLOs scoring above threshold levels, aside from relatively high rankings for AoW2. This does not reflect a lack of need, but indicates that, given limited resources, these regions may currently present fewer opportunities for high-impact engagement for the Program.

5.2 Final decisions on the Program's direction and design

For AoW1, the final programmatic decisions were shaped to highlight its unique value as a pivotal area of work cutting across the three food systems, positioning it as a global convening and synthesis platform. AoW1 aims to bridge knowledge gaps and foster transformative learning across these food systems and diverse regions. The inclusion of areas like CWANA, despite scoring lower in knowledge-driven prioritization, underscores a commitment to global inclusivity and comprehensive synthesis. These decisions were further refined through a comparative advantage analysis, ensuring CGIAR's strengths are leveraged to maximize impact on HLOs. Additionally, discussions across AoWs ensured alignment and strategic coherence, reinforcing AoW1's role as a cornerstone in delivering unified and impactful outcomes.

For AoW2, programmatic decisions were made by triangulating the prioritization data with partner country demand, donor interest, strategic alignment across the AoW, and existing implementation capacity. For example, while WCA scored moderately in the analysis, strong institutional partnerships and urgent on-the-ground needs, particularly in conflict-affected areas—justified its inclusion in several programmatic strands. This is further supported by the significant earmark donor allocation (Foreign, Commonwealth & Development Office (FCDO) and Swiss Agency for Development and Cooperation (SDC)) and specific commitment to AoW2 in FCAS. Although an enabling environment in these settings may be limited, the potential for impact remains strong.

For AoW3, five cities were strategically selected to serve as primary targets. These are Nairobi, Accra, Lima, Dhaka and Manila. They represent a diverse spectrum of unique urban food systems challenges, offering distinct and valuable entry points for CGIAR innovations to contribute

impactful and context-specific solutions. These decisions were guided by engagements with local municipalities and a diverse range of other stakeholders.

For AoW4, key considerations included the vulnerability of island populations, strong in-country programs, regional partnerships, and donor priorities—such as Australia’s earmarked W2 funding for the Pacific. Due to underrepresentation in prioritization model inputs (e.g., small populations and economies), as well as CA and co-design with partners, scores were adjusted accordingly. This ensured that the Pacific remains a priority region, with significant AoW4 research efforts focused on achieving HLOs in the Pacific Islands. Conversely, CWANA and LAC, which scored consistently low in the prioritization analysis, have been deprioritized due to both limited donor interest and low thematic alignment. However, targeted engagement may continue where donor or policy windows emerge (e.g. emerging opportunities in the Caribbean with AoW4 on island food systems).

The final Prioritization also ensured complementarity with investments in other SPs (e.g., the [Policy Innovations SP](#) and [Gender Equality and Social Inclusion Accelerator](#)), avoiding duplication and enabling co-funding in shared geographies such as ESA and SA. Equity and inclusion were cross-cutting considerations, especially when prioritizing interventions in fragile contexts where gender equality is most at risk, (AoW2), urban informal settlements (AoW3), and youth-oriented innovation hubs in the Pacific (AoW4). Regarding MELIA, most of the studies focus on regions (ESA, SA, and WCA), and AoWs, where major donor investments are focused.

Overall, the Prioritization exercise helped to discontinue and consolidate HLOs. For AoW1, HLOs were consolidated from five to three, with the highest investment in HLO1.3, followed by 1.2, then 1.1. For AoW2, HLOs were also consolidated to three only, whereas most of the investment is going to HLO2.2, followed by HLO2.3 and 2.1. For AoW3, the Prioritization process was instrumental in pinpointing five HLOs which represent thematic areas of strategic focus. For AoW4, the HLOs remained representative of research demand with Pacific Island partners and the priorities where the Program is directing research investment.

Refer to the Prioritization attachment for more information.

6. Alignment of W3 and bilaterally funded work

A total of eleven bilateral projects—representing the largest 80% (by value) of Center-led bilateral projects—have been mapped to the Food Frontiers and Security Program. Their combined mapped budget for 2025 is USD 7.6 million. These span AoW2 and AoW3; we are aware that there are smaller bilateral projects in the Centers, across the AoWs, below the 80% threshold.

An example of a strong cross-portfolio linkage in AoW2 is [International Water Management Institute's](#) (IWMI's) bilateral [Jahez project](#) (funded by FCDO) in Jordan and Lebanon. The project resulted from the work in the Fragility, Conflict, and Migration Initiative and is tied to the Program's carryover theme of building resilience in communities hosting displaced people. Another core example, supporting AoW2's modeling and foresight work, comprises two bilateral grants: one from Google on Artificial Intelligence (AI)-driven crisis and food security analytics and another from the [European Commission](#), funding Phase 4 of the [Food Security Portal](#) and strengthening anticipatory action and crisis response through real-time food crisis monitoring tools.

Additional bilaterally funded projects commissioned through [International Food Policy Research Institute](#) (IFPRI) complement the W1/W2-funded Program by enhancing its contextual relevance, methodological diversity, and policy engagement. Projects like the [Intimate Partner Violence \(IPV\) Research Collaborative](#) and [Strengthen PSNP Institutions and Resilience II \(SPIR II\)](#) deepen the Program's integration with national systems and policy processes as well as core gender equality and social inclusion objectives. [Yemen's food security project](#) supports capacity-building and evidence-based planning—reinforcing the Program's applied and adaptive approaches. The [Cash for Work](#) evaluation expand analytical depth in fragility and resilience. Together, these investments fill geographic and thematic gaps, strengthen implementation linkages, and enhance the Program's ability to deliver on its ToC and partner priorities.

In AoW3, the [International Potato Center's](#) (CIP's) 1572-MIPO project in Peru is promoting Yacon, a root vegetable with scientifically demonstrated benefits for lowering blood sugar levels. Meanwhile, CIP's 1529-UNGP (World Bank Group) project in Madagascar is reducing food loss in the sweet potato value chain and creating quality employment for women and youth in the food processing sector, in collaboration with private enterprises.

However, the mapping also reveals critical gaps. AoW1, the Foresight and Synthesis Hub, is underrepresented in bilateral funding despite its central role in Program integration, cross-regional learning, and adaptive management. Additionally, AoW4 on island food systems is a new and growing component of the CGIAR portfolio highlighting the opportunity for growing the bilateral investment in island regions through this framework of engagement. Addressing these opportunities through targeted resource mobilization, co-investment mechanisms, and strategic engagement with funders will help the Program fully realize its ambition for high-impact, inclusive, and scalable solutions across all four AoWs.

7. Plan of Results and Budget

Background The 2025 PORB builds on the partnership dialogues and the Prioritization and CA analysis carried out during the Inception Phase. However, it is also influenced by past structures and allocations; during the CGIAR Portfolio design in 2024, the CGIAR sought to retain at least 80% of the average 2022-2024 Initiative budgets in acknowledgement of continuity of existing engagements. The Food Frontiers and Security Program evolves three 2022-2024 Initiatives into four new AoWs and is therefore influenced by these legacy decisions. The budgeting for the 2025 PORB departs from this legacy to some effect driven by Prioritization and the CA analysis, but also with adjustments based on donor designations in Window 2 (W2) and allocations in Window 1 (W1).

Program allocations The Program received a total of \$8.89 million whereof \$4.88 million was W1/W2 at program level (\$2.51 million W2 program designation), and \$4.01 million was W2 designations at AoW level: AoW2 received \$2.12 million in W2 designation from FCDO (United Kingdom) and SDC (Switzerland), and AoW4 received \$1.89 million W2 designation from Australian Centre for International Agricultural Research (ACIAR) (Australia). The total W2 designation of \$6.52 million (73% of the total) indicates strong alignment with the priorities of several funders. The W2 funds helped strengthen the priorities of the overall program. But the way W2 mapped into an already significantly reduced overall allocation challenged the Transitional Leadership team in prioritization and consolidation around an already underfunded ToC. Specifically, the designation to AoW4 increased its budget to nearly 2.5 times its initial level without any increase in the overall Program envelope. Rather, the Program received a decrease like that of other programs in the Portfolio, naturally causing significantly larger reductions to budgets of AoW1, AoW2, and AoW3. For the case of AoW2, the reduction was partly compensated by W2 funds, but these did not reach the level of its initial budget. Recognizing that the W2 designation in AoW4 was higher than the legacy pooled funds from previous Initiatives allocated in 2024, and given AoW1 and AoW3 did not receive W2, AoW4 rescinded its W1 allocation to AoW1 and AoW3, to secure the Program's coherence and sequencing of synthesis from the three frontier food systems.

Nonetheless, AoW1, 2, and 3 remain significantly underfunded—particularly AoW1 and AoW3. The Program is therefore working to address this through coordination with other Programs and targeted fundraising. This includes a mix of pursuing core CGIAR donors to showcase achievements and potential (e.g., to highlight value-add of the program where it might not have been clear to date), informing and advising development banks and governments on investments in those thematic areas as well as pursuing new donors and seeking smaller grants to co-support key research and scaling activities. Partnerships with private sector actors whose values and priorities align with CGIAR for co-investment opportunities in specific thematic areas are also being explored. We are additionally reaching out to other SPs with linked HLOs to discuss how our programs might increasingly work together, particularly, in 2026, to generate more synergies.

AoW1 Future Food Lab The AoW1 PORB reflects a nearly 46% reduction from February 2025 W1/W2 confirmations yet strategically reallocates funds to sustain and amplify its role as the Program's synthesis, learning, and innovation hub. AoW1 integrates evidence, insights, and innovation across the Program, enabling cross-center collaboration and alignment. HLO1.3 commands the largest share at \$0.35million, underscoring the priority placed on entrepreneurship support and scaling high-potential innovations with select innovators, entrepreneurs, public actors, and youth in resource-constrained food systems. HLO1.2 receives \$0.12million focusing on co-developing knowledge products, action plans, and investment priorities, directly strengthening stakeholders' capacity to deliver resilience-building interventions in frontier geographies. HLO1.1 is allocated \$0.06million to support the generation of tools and evidence for policymakers and

partners to navigate emerging trends and scenarios. Collectively, the PORB reflects AoW1's integrative role in advancing adaptive, forward-looking, and system-wide solutions.

AoW2 Fragile and conflict affected food systems AoW2 experienced a 30% reduction relative to initial February 2025 W1/W2 confirmations—smaller than some other AoWs, reflecting the Program's and donor partners' recognition of its centrality to the Frontiers agenda, but still far smaller than its planned size. AoW2 directly addresses the three faces of crises and conflict: preventing and anticipating (HLO2.1), responding (HLO2.2), and recovery and stabilization (HLO2.3)—in some of the world's most challenging food system environments. HLO2.1, which generates evidence for anticipatory action and disaster risk reduction, receives 1.05 million. HLO2.2, aimed at strengthening the effectiveness of ongoing humanitarian responses, receives 1.24million. HLO2.3, focused on recovery and resilience in underserved fragile contexts, receives 1.85 million. These investments support a coherent and scalable body of work that is both demand-driven and grounded in deep partnerships with frontline actors. The AoW also leverages \$2.12 million in W2 designations from FCDO and SDC, underscoring the importance placed on fragile and conflict-affected settings within the broader food systems transformation agenda. Indeed, the SDC earmark reflects a doubling over the previous year's allocation.

AoW3 Urban food systems The AoW3 budget allocation of \$1.33 million is distributed across 5 HLO's which aligns key priorities in the target cities and strategic CGIAR entry points for transformative urban food systems. Building on CIP's foundational management role under the [Resilient Cities Initiative](#), CIP continues to hold significant oversight on strategic direction and overall delivery, leading AoW3 with [World Vegetable Center](#). HLO3.1 funds [International Institute of Tropical Agriculture](#) (IITA) to deliver safe, nutritious and climate resilient urban food production, in collaboration with World Vegetable Center, at a budget of \$0.26 million. HLO3.2 receives \$0.16 million for IWMI to deliver on improved methods of organic waste recycling and wastewater reuse for urban production. HLO3.3 enhances inclusive informal food markets that effectively create quality jobs for young women and men and deliver safe and healthy foods in urban environments. ILRI is leading the work in collaboration with CIP, [International Rice Research Institute](#) (IRRI) and the [Alliance of Bioversity and the International Center for Tropical Agriculture \(Alliance of Bioversity and CIAT\)](#). HLO3.4's allocation \$0.20 million focuses on interventions for improving diet quality for urban residents, led by IFPRI. These efforts are all strategically guided by HLO3.5 led by CIP on urban food systems governance, targeting policy makers and key stakeholders to guide interventions aimed at strengthening resilience in urban food systems. Comprehensive allocation of resources underscores an integrated approach to building resilient urban food systems.

AoW4 Island food systems The AoW4 allocation is distributed across the five HLOs to match Pacific partner expectations and priorities. AoW4 is a new direction for the CGIAR portfolio and builds on [WorldFish](#)'s 40-year partnership program in the Pacific. WorldFish therefore holds significant oversight and manages contractual arrangements with partners. HLO4.1 is the biggest investment at \$0.44 million. This includes funds to World Vegetable Centre and CIP to seed a deliberate engagement of [Genebanks](#) work in the Pacific with evolving Pacific partnerships in this thematic area. HLO4.2 receives \$0.24 million to support efforts on nutrition, Indigenous foods and school meal programs. HLO4.3 is allocated \$0.29 million to engage [Solomon Islands National University](#) and [Kastom Gaden Association](#) to form collaborative efforts at the [Nusatupe innovation hub](#) for island food systems with a strong focus on youth internship, living labs and Indigenous knowledge for island futures. This work is linked with HLO4.4 which receives \$0.30 million and includes an allocation to IWMI for emerging work on circular economy at islands, transferring technologies from urban food systems. Finally, HLO4.5 receives \$0.34 million, whereof the Alliance of Bioversity and CIAT holds a \$0.12 million allocation for integration of climate action in ACIAR

funded bilateral projects that engage over 10 partners in the region. This HLO is a fundamental element of the ambition of a generalizable framework for diagnosing island food system resilience with [Australian National Centre for Ocean Resources and Security \(ANCORS\)](#) and the [SPC](#).

Refer to the PORB attachment for detailed information.

8. Cross-Portfolio linkages and geographic coordination

8.1 Cross-portfolio linkages

Scaling for Impact Program

The Scaling for Impact (S4I) Program is a critical partner in advancing the Food Frontiers and Security Program by adapting proven scaling approaches to fragile, urban, and island food systems. In collaboration with S4I, the Program catalogues CGIAR and external innovations and applies tools like joint [Innovation Package and Scaling Readiness \(IPSR\)](#) workshops to develop context-specific scaling pathways and strategies. This work contributes to the Program's HLOs 1.2, 1.3 and IOs 1.2, 1.3 by enabling innovation uptake in resource-constrained areas. Through AoW5, they leverage the [CGIAR Scaling Hub](#) in Nairobi to deepen learning on scaling in fragile settings and document successful scaling cases through joint working groups and publication. Joint work with AoW3 focuses on scaling circular economy solutions in urban systems (HLO3.2) through ecosystem analysis and stakeholder engagement, and the outputs from [Nusatupe innovation hub](#) for island food systems (HLO4.3) features within this interaction too. The partnership also documents lessons from agribusiness acceleration to refine entrepreneurship support models for fragile geographies, enhancing innovation delivery and market resilience.

Policy Innovations Science Program

The Policy Innovations SP partners with the Food Frontiers and Security Program in addressing governance and policy challenges across fragile, urban, and island food systems. Their collaboration contributes to HLOs 1.1-2, 2.2-3, 3.2-5 and 4.1 as well as IOs 1.1, 1.2, 3.2, 3.3, 3.4 and 3.5, generating evidence, tools, and insights for better policy and resilience. In fragile settings (AoW2), the focus is on co-developing modeling and foresight tools, analyzing social protection programming, and conducting political economy analysis for crisis response. In urban areas (AoW3), they co-produce evidence with networks, support decision-tool development, and build institutional capacity for inclusive governance. The Food Frontiers and Security Program will organize joint knowledge and capacity sharing events with the Policy Innovations SP engaging municipal governments as well as with city networks. This partnership ensures policy innovation is grounded and supports effective action in complex, frontier systems.

8.2 Geographic coordination

The [Climate Action SP](#) and the Food Frontiers and Security Program's AoW2 coordinate geographically in a deliberately complementary—not co-located—manner. While “coordination” often implies working in the same places, in this case it means covering distinct but strategically aligned geographies to maximize responsiveness to stakeholder needs, as well as efficiency and impact. The Climate Action SP focuses primarily on less fragile contexts, while AoW2 leads work in FCAS, including those affected by displacement and migration. This coordination is carried out through the joint development and use of two key tools—the [Climate Security Observatory](#) and the [Climate Security Programming Dashboard](#)—launched in 2023 and covering 12 countries. These platforms form the backbone of our collaboration—enabling evidence-based prioritization, programming, and financing decisions across a wide geographic footprint. Together, we use them to inform national strategies (e.g., Zambia, Kenya), shape regional responses (e.g., Central America, Horn of Africa), and support operational planning by partners such as UNHCR and WFP (e.g., in Mali, Burkina Faso, Niger, Mozambique, and Jordan).

In partnership with the [Anticipation Hub](#), we are also expanding the Early Warning, Early Action and Early Finance ([AWARE](#)) platform to strengthen early warning systems. Here again, the Climate Action SP contributes technical capabilities, while AoW2 ensures integration and uptake in fragile contexts where warnings often fail to reach the most vulnerable communities. By working in different types of settings but using shared tools and aligned approaches, the two Programs achieve

operational efficiency, avoid duplication, and generate broader, more context-responsive insights than either program could produce alone, delivering greater value to partners and stakeholders across the CGIAR portfolio. Together with The Climate Action SP, we will work with CGIAR Country Conveners as on-the-ground brokers to facilitate alignment with national priorities, partner demands, and stakeholder needs. In Kenya, for example, we will leverage the CGIAR Country Convener to align climate resilience and food security efforts, foster coordinated engagement with government, local partners, and development actors, and enhance the collective impact of our interventions.

During the Inception Phase we developed a new opportunity with the [Genebanks](#) specifically focused on the Pacific (AoW4). In this planned work AoW4 with its networks and partnerships provides the framework and coordinating function for broader CGIAR engagement.

9. Risk management

Please refer to the following [link](#) with the Program/Accelerator's full risk register. This link is open to ISDC colleagues and others will be granted access, on request.

#	Category	Title	Description	Current Risk Level	Target Risk Level	Actions /Controls to manage risks
3	Funding	Unpredictable/ inadequate funding	Funding uncertainty (inflation, cuts, etc) reduces our ability to develop long-term partnerships, while sudden reductions affect our credibility with partners. Existing partners unable to meet Program requirements due to their budgetary constraints and commitments.	20	12	(1) Diversify Funding Sources (2) Develop Contingency Plans (3) Co-Finance with Partners (4) Strengthen Financial Forecasting and Pipeline Tracking (5) Communicate Transparently with Partners (6) Build a Compelling Value Proposition
4	Funding	Catastrophe diverts resources	Increased frequency and severity of climate-related events (e.g., flood, drought) and conflict, overwhelming food systems where the Program works, or diverting financial and human resources for response.	16	12	(1) Integrate Risk-Responsive and Adaptive Programming (2) Establish Business Continuity and Crisis Management Protocols (3) Strengthen Early Warning and Financial Scenario Monitoring (4) Diversify Funding and Maintain Emergency Flex Pools (5) Build Resilience into Program and Institutional Design (6) Forge Humanitarian and Multisectoral Partnerships (7) Design Interventions that are Flexible and Shock-responsive
5	Talent	Changes in the Operating Environment	Frequent internal restructuring, leadership transitions, and shifts in strategic direction create uncertainty, disrupt workflows, reduce staff morale, and delay decision-making.	15	12	(1) Establish strong Internal Communication System (2) Develop a Change Management Strategy (3) Document Core Processes and Knowledge (4) Strengthen Decentralized Decision-Making (5) Align Program Design with Institutional Strategy (6) Regular Risk Review Meetings

1	Scaling impact	Inadequate capacity to implement innovations	Government and non-government partners have limited financial and technical capacity. Investment in capacity sharing may not be adequate to enable partners to implement innovations needed to improve food security/resilience.	12	9	(1) Leverage Local Institutions (2) Conduct Capacity Gap Assessments (3) Promote Co-Funding and Resource Mobilization (4) Use Innovation Readiness Metrics
2	Partners and Partnerships	Weak governance among implementation partners	A major dimension of food system fragility is weak governance, which may increase the exclusion of those with limited resources, especially women and youth, undermining our ability to target these populations.	12	8	(1) Facilitate Policy Dialogue and Coordination (2) Use Social Accountability Tools (3) Build Institutional Governance Capacity

10. Addressing feedback on select topics

There is a lack of information on how water systems are covered in the Portfolio.	Water systems are integrated across all AoWs in the Food Frontiers and Security Program, with particular emphasis on climate-smart innovations, circular bioeconomy solutions, disaster risk reduction, and digital tools for resource management. The Future Food Systems Lab (AoW1) plays a central role in synthesizing water-related insights across fragile, urban, and island contexts. In FCAS (AoW2), the Program supports anticipatory action and disaster risk reduction through tools that address flood and drought events. For example, digital twin modeling is being piloted in Dolo Ado, Ethiopia to simulate water system dynamics and inform humanitarian planning. In Eastern Sudan, the Program is advancing flood and agricultural water management solutions, while in Jordan, water risk profiles are being developed for refugee camps to guide resilience-building investments. In island systems, safe wastewater reuse and aquatic food system resilience are key priorities. Additionally, AoW1 and AoW2 are co-producing a learning series with the Fragility, Conflict, Livelihoods and Water Research Group to promote applied learning on water governance, conflict-sensitive resource management, and practical, scalable solutions for fragile and underserved settings. These efforts will be further elaborated in the Inception Phase to strengthen the visibility and coherence of water systems research across the Portfolio.
There is a lack of information on how climate mitigation is covered in the Portfolio.	The Program focuses more on climate adaptation than mitigation. However, AoW1 explores circular bioeconomy models and innovations (e.g., vertical farming, waste-to-protein systems) that can contribute to mitigation. Further elaboration of mitigation strategies, including GHG emissions baselining and nature-based solutions in urban and island settings, will be prioritized during the Inception Phase. Under AoW3, HLO 3.1 focuses on Safe Urban Food Production which promotes local food production in urban gardens. This significantly reduces the emissions from long-distance transport, a major contributor to urban food systems' carbon footprint. Additionally, HLO 3.2 on Circular Bioeconomy which focuses on organic waste recycling and safe wastewater reuse contribute towards diverting organic waste from landfills and reducing emissions from freshwater treatment respectively, both key climate mitigation strategies. In AoW4, which focuses its work on the small island states in the Pacific, climate work focuses on adaptation and climate risk analyses. Nevertheless, work on circular production systems and small-scale fisheries are entry points for mitigation (reduced fuel consumption, low technology practice) that will be included in HLO 4.3
There is a lack of information on how tradeoffs and synergies in mixed systems are covered in the Portfolio.	The Program's transdisciplinary approach anticipates mixed systems challenges, particularly in urban and peri-urban environments. AoW1 (Future Food Systems Lab) is tasked with scenario planning and foresight analysis that considers such tradeoffs. In addition, in AoW2, we are identifying climate change – conflict hotspots that help identify synergies for climate-smart agriculture and peace-building activities. Additionally, AoW2 is expanding our use of big data to better highlight where investments in crises prevention, response, and recovery are likely to be highest value – promoting more effective policies in an environment beset with numerous investment tradeoffs.

There is a lack of information on how environmental health and biodiversity is covered in the Portfolio.	The Program references environmental degradation and ecosystem health (e.g., urban food waste recycling, soil microbiomes, restoration of indigenous aquatic systems). However, biodiversity and environmental health outcomes are not deeply elaborated. This will be addressed through targeted research questions under AoW1 and AoW4 and alignment with the Environmental Health & Biodiversity Impact Area.
There is a lack of information on how orphan and opportunity crops and pulses are covered in the Portfolio.	While not central, the Program acknowledges local crop systems in island contexts and urban agriculture, where pulses and nutrient-rich crops can be emphasized. AoW3 and AoW4 will explore this further in collaboration with the Better Diets and Nutrition Program and NARES partners. Opportunity exists to include neglected crops in food environment and resilience-building strategies in the Inception Phase. AoW3 continues to actively promote urban production of a wide range of indigenous vegetables in Asia and Africa under HLO 3.1 through training in learning institutions (schools and colleges) and community gardens. For example, in 2023, AoW3 introduced 68 varieties of 31 types of indigenous food crops in urban gardens in Manila.
There is a lack of information on how youth and social inclusion (that does not relate to gender) is covered in the Portfolio.	The Program highlights youth inclusion, especially in job creation, agri-entrepreneurship, and urban innovation hubs. Youth engagement is prominent in AoW3 (Urban Food Systems) and AoW4 (Island Food Systems). AoW2 further recognizes youth as a critical group that can either help perpetuate or reduce fragility and conflict, requiring tailored programming and interventions – and addresses issues of migration (dominated by youth). However, broader social inclusion is less developed and will be addressed via GESI-responsive programming and metrics during the Inception Phase.
There is a lack of information on how capacity sharing (as it relates to internal learning and decolonization of research) is covered in the Portfolio.	Capacity sharing is integrated in each AoW. AoW1 promotes co-creation with local partners, and the Program builds on long-term partnerships with NARES and local innovators. Deep commitments to decolonizing research—such as equitable authorship and local leadership in agenda-setting—will be strengthened as part of the Capacity Sharing Accelerator. Under AoW2, we will implement the Frontiers Learning Exchange Program in partnership with IWMI's Fragility, Conflict, Livelihoods, and Water Research Group to support knowledge capture and to cultivate innovative, practical solutions in FCAS. We will also deploy a 'From Fragility to Stability' webinar series as part of IFPRI's Policy Research Seminar Series. Finally, across AoWs, we have set targets for capacity development—ensuring that the tools and innovations we develop can scale and yield long-term benefits. AoW2 also launched the Climate, Peace and Displacement (CPD), which specifically builds on the localization agenda to i) take CGIAR science and expertise to local and international humanitarian and development actors ii) creates spaces for dialogue and exchange between food, land and water systems science and humanitarian policy and iii), directly supports science in Africa through mentorship programs with individuals, Centres of excellence and regional networks.

Appendix 1: Addressing specific feedback

FCDO: “AoW2 is of particular interest. We welcome the CG’s efforts to tackle this global research gap - in particular, evidence on what works to strengthen food security and resilience in FCAS. What role can this SP play to support other SPs and draw together evidence on what works in FCAS?”	This is a critical value proposition of AoW2 of the Frontiers SP—expanding the reach of CGIAR to studying topics of fragility and conflict (not only working in FCAS settings) as well as increasing the efficiency of CGIAR investments carried out by other SPs/ Accelerators in FCAS settings. For instance, With the Climate Action SP, we are co-developing tools like the Climate Security Observatory and AWARE platform, enabling conflict- and climate-sensitive programming in fragile settings. We are collaborating with the Digital Innovation Accelerator to develop a digital twin of a watershed in Ethiopia to support humanitarian response in refugee host communities. And with the Gender Equality and Social Inclusion Accelerator, we are advancing gender-responsive crisis prevention and recovery—analyzing how conflict, climate shocks, and migration affect women’s livelihoods and agency, and testing interventions to empower women as crisis responders and leaders in fragile contexts. With the Policy Innovation SP, we are convening dialogues on core topics of policy interest (e.g., an April 2025 event in Nairobi on Sudan’s ongoing crises – “Two Years of Crisis: Impact and Pathways to Recovery and Resilience” – as well as an event on youth). And we are working with the Scaling 4 Impact SP to support scaling in and to FCA settings. During the Inception Phase, we have connected with these and other SPs/As to plan joint work and dissemination and magnify our collective impacts whilst establishing best practices for this type of work in the CGIAR. Finally, the Frontiers Learning Exchange (FLEX) will initiate a community of practice for shared learning and innovative problem-solving in FCAS environments.
ISDC: “The specific challenges facing each of these distinct at-risk areas are vastly different, limiting the potential gains from housing all three under one program.”	During the Inception Phase the program has sought to clarify that in fact these systems share characteristics and challenges (see Figure 2 and Section 1.3). This was well received at Science Week in Nairobi where the Program enjoyed novel engagement from funders and partners because of the increasing awareness of the novelty and innovation in framing that the program represents. Within the framework of the program there is still significant scope for tailored research, alongside systematic exchange and opportunities for transfer of technologies. For example, during the Inception Phase we have allocated resources for transfer of circular economy research tools and technologies previously applied in urban food systems to island food systems. This new iteration of programming should make the case clearer for housing these three frontier food systems together.

ISDC: “The SP contemplates a significant investment in research in FCA areas, where CGIAR has not operated previously...AoW2 involves a new focus for the CGIAR that may involve substantial risks... The eventual effectiveness of the SP could be hampered by the need to establish capacity and inherent costs and risks involved.”	AoW2 on FCAS is a natural extension of the already successful FCM Initiative that delivered targeted research and partnerships in FCA settings. Building on the foundation laid by FCM, this AoW has mobilized the same cadre of CGIAR scientists and a revised (for enhanced impact potential) set of partners and geographies to co-produce context-specific innovations and evidence to strengthen food system resilience amid conflict and climate shocks. Formal collaborations with trans-national actors like WFP, UNHCR, and IOM, as well as NGOs (e.g., ActionAid, World Vision) and government counterparts in over 10 countries, are already underway. Recognizing that fragility is intensifying even in traditionally stable CGIAR regions, AoW2 (as well as AoW1) are also advancing tools and innovations to bridge the humanitarian-development-peace nexus—directly addressing partner demands. These early efforts demonstrate CGIAR’s comparative advantage and strong institutional commitment to effective operation in high-risk food systems. SDC specifically welcomes the nexus approach in FCA settings and has thus earmarked AoW2, which aligns with SDC’s own approach and priorities.
ISDC: “The failure to engage with the typical thinness of adequate, existing secondary datasets and the difficult challenges associated with generating new and credible primary data for impact evaluation and other purposes in FCAS”	Food systems in FCA areas are indeed marked by data scarcity and challenges to conventional data collection. During the Inception Phase, these constraints have been addressed through several adaptive approaches. First, the SP has expanded work on and capacity in modeling and remote sensing, partnering with the Food Security Portal (FSP) team as well as Google to model, predict, and better understand the likely impacts of shocks and crises. Second, we have more broadly increased our reliance on novel data sources including those stemming from remote sensing, phone-based surveys to reach insecure or inaccessible areas, and external databases such as ACLED. We have also established flexible partnerships with local organizations and UN agencies to co-generate novel data. These strategies have already enabled the production of credible, policy-relevant evidence where traditional methods would not be feasible.
ISDC: “The lack of well-supported hypotheses and credible research literature references	We are grateful that the Inception Phase has afforded us opportunities to articulate our research questions, theories of change, and assumptions therein. The value added and overall merit of the Food Frontier and Security Program at this stage is grounded in the substantial body of research, literature reviews, and partner engagement conducted during the 2022–2024 CGIAR Initiatives—each of which was itself deeply rooted in the legacy of the CGIAR Research Programs (CRPs). While space and time constraints in this Inception Report limit our ability to detail these foundations, they have directly informed the design of our work plans, theories of change, and

makes it very difficult to assess originality and likely value added and overall merit of the SP.”	research priorities. The Program builds on this robust evidence base to ensure that its contributions are both original and impactful. As implementation progresses, we will continue to draw on this foundation to deliver high-quality, context-relevant research that provides strong value for money for donors and actionable insights for uptake partners. This is not a static or linear process—we believe that implementing an impactful science program requires a dynamic approach, one that continuously assesses existing and emerging knowledge, and actively engages with partners to ensure that what we produce remains relevant, useful, and appropriate.
ISDC: “Gender aspects at institutional/ partner level need to be highlighted, i.e., demonstrating efforts to ensure that the SP’s team meets CGIAR’s gender target of a min. of 40% women in professional roles.”	This is a critical point. We flag that of 10 individuals serving either as interim director, interim deputy director, or interim area of work lead/ co-lead, 4 are women. Further, amongst these 4 women, 2 are currently serving as interim deputy directors. This provides core balance. Additionally, within each area of CGIAR work team, leads are making efforts to ensure researchers from the global south in addition to women are given opportunities as key technical focal points and project leads. Finally, we are ensuring that female voices are among partner teams by selecting partners who allow for balanced voices on teams working with us.
ISDC: “Providing a more comprehensive and thought through risk framework, with explicit mitigation actions and contingency plans for possible disruptions and unintended consequences would significantly enhance the robustness of the proposal.”	We fully agree that risk management is critical, particularly in fragile and frontier settings, and have already identified the risks facing the SP as well as designed concrete mitigation strategies across AoWs. In AoW2, which operates in FCAS, risk to staff and human subjects is being managed by partnering with organizations experienced in such contexts, aligning with UN security protocols, applying rigorous IRB and ethical standards, and using remote data collection methods (e.g., phone surveys, satellite imagery) where necessary. In AoW3, operating in complex urban environments, risks from unrest are being addressed through advance risk assessments, real-time monitoring, and engagement with politically neutral, technically grounded local partners. In AoW4, we are managing risk through embedded local staff, long-term partnerships, and adaptive field protocols. We are drawing on experience operating during past civil conflict in Solomon Islands, co-designing activities with national and local partners to ensure neutrality and incorporating contingency planning for climate-related disruptions such as cyclones and flooding which occur annually.

ISDC: “Given the CGIAR experience with participatory approaches, the limited reference to and inclusion of farmer organisations and other grassroots perspectives is surprising”	AoW2 is working with farmers directly as well as civil society organizations (formal and informal) in a broad set of countries. For example, in Mali we are working with over 3,000 local leaders across all regions of the country (including women leaders; youth leaders; and formal and informal local leaders) to better understand the impacts of shocks and crises and which solutions are most effective to anticipate risks. In Nigeria, we have trained women’s organizations in 450 communities across three states to better engage with their leaders and demand investments that support their livelihoods and incomes. And in Ethiopia we are working with local community leaders to improve targeting of social assistance programs and rebuild trust in community leaders. AoW3 is actively engaged with community level actors involved in urban and peri-urban food systems, including small scale producers, informal market vendors, women, youth and community groups. In Kenya and Philippines, the Vendor Business School empowered informal market vendors, mainly women, with business development and food safety skills. The work in Lima, Peru showcases direct grassroots engagement through the food rescue initiative where surplus produce from farmers and markets is collected and channeled to underserved communities. This involves working with local market actors and community groups to improve food access for vulnerable communities. In Dhaka, Bangladesh, AoW3 established four farmers markets with the aim of shortening the value chain by providing farmers with a platform to market their products directly to consumers. These farmers markets are managed collaboratively by the local government and the farmers groups, AoW3? AoW4?
ISDC: “A more explicit identification of potential competitor organizations and whether CGIAR is the best among the group to take such tasks is warranted. The identification of potential partners is important, but just as important is to identify others working in the space that are not likely to be partners.”	The Food Frontiers and Security Program’s comparative advantage analysis explicitly considered the broader ecosystem of actors working in fragile, urban, and island food, land, and water systems. This included identifying both complementary partners and other organizations operating in similar spaces that may not be direct collaborators. The analysis highlighted specific sources of CGIAR comparative advantage in frontier geographies: our embedded partnerships, on-the-ground presence in many settings, systems research heritage, and public goods mandate. These enable CGIAR to operate effectively in politically sensitive and data-scarce environments where many cannot. While the SP actively collaborates with a wide range of partners—including UN agencies, humanitarian organizations, governments, and local innovation hubs—it also recognizes that other actors such as the World Bank, FAO, WFP, UNHCR, IFRC, and various INGOs (e.g., CARE, ActionAid, World Vision) engage in overlapping domains. We do not seek to duplicate their efforts but rather complement them by contributing rigorous, systems-oriented research and innovation design that these operational actors can apply and scale. This awareness of the broader landscape has shaped the SP’s design, including decisions to focus on areas where CGIAR’s scientific capabilities and long-term partnerships offer the greatest added value. It has also informed the Program’s division of labor—where CGIAR leads on diagnostics, innovation, and evidence generation, while other actors lead on delivery, policy uptake, or humanitarian response. This strategic clarity ensures that the Program is not only well-positioned but also well-differentiated within a crowded field
ISDC: “Area of Work 1 (The Lab) is very	AoW1 adds value to the SP by drawing on CGIAR’s transdisciplinary expertise across centers and food systems, its embedded presence in national institutions, and its robust global-to-local partnerships. These capabilities position the Future Food Systems Lab to deliver

important to the Program and has substantial potential to add value. However, added details are needed to determine whether this potential is realized.”	effectively across all High-Level Outputs (HLOs). Under HLO 1.1, the Lab will leverage CGIAR’s longstanding engagement with policymakers to shape strategies in complex, data-scarce environments. HLO 1.2 will synthesize global evidence from fragile, urban, and island food systems and collaborate with local stakeholders to generate context-specific, actionable insights for designing resilience-building interventions. For HLO 1.3, the Lab will build on proven models - such as the Stability and Peace Accelerator - to identify, support, and scale local innovations that address challenges in frontier food systems. Together, these strengths ensure that AoW1 is well positioned to turn its potential into measurable and equitable impact.
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